A Novel Dense Generative Net Based on Satellite Remote Sensing Images for Vehicle Classification Under Foggy Weather Conditions

Jianjun Yuan, *Member, IEEE*, Tong Liu, Haobo Xia, and Xu Zou

***Abstract*— Accurate vehicle-type classification plays a crucial role in the development of intelligent transportation systems. Recently, several deep learning models have been proposed to utilize satellite remote sensing images for vehicle-type classification. However, conventional neural network models often have limitations when dealing with remote sensing images, such as adverse weather conditions, as well as the extremely low resolution of remote sensing images containing small objects like vehicles. To enhance the vehicle-type classification capability in complex environments, this article develops a novel deep learning framework called Dense Generative Net (DGNet). DGNet consists of three components: feature layer, generation layer, and dense feature fusion layer. The feature layer employs large convolutions to establish a broader receptive field, enabling it to capture more effective global feature information. The generation layer is based on a super-resolution (SR) network, which is designed to generate high-resolution feature information. The dense feature fusion layer performs the final classification by integrating the outputs from two upstream branches and combines the feature information obtained from the feature layer and the generated high-resolution features information from the generation layer, enabling comprehensive and robust classification of vehicle types. To evaluate recognition capability, vehicle data from multiple regions and diverse environmental conditions are utilized, including four different weather conditions. The experimental results demonstrate that DGNet exhibits remarkable vehicle-type recognition capability, with minimal degradation even under heavy foggy weather conditions. The effectiveness of each module and its impact on the overall performance have been verified through ablation experiments.**

***Index Terms*— Dense Generative Net (DGNet), Foggy weather conditions, satellite remote sensing, vehicle classification.**

I. Introduction

VEHICLE-TYPE classification is a crucial and challenging topic in the field of intelligent traffic systems. It includes predicting time-series traffic information [1], managing transportation facility usage [2], conducting traffic analysis, determining fuel types, and analyzing driver behavior. Numerous methods have been developed to enhance the accuracy of vehicle classification using either specific hardware components (such as sensors) or ordinary digital cameras, and these methods have yielded promising results [3]. Vehicle classification using image processing approaches can be divided into two main categories: appearance-based methods and deep learning-based methods. In recent years, the use of satellite remote-sensing images for intelligent traffic systems has become more widespread due to the availability of unmanned aerial vehicle (UAV) surveillance and satellite remote-sensing data. In this article, we focus primarily on deep learning-based methods for vehicle classification using remote sensing imagery.

Remote sensing imagery can cover a wider area, enabling analysis and management of large-scale vehicle data, which is particularly important for modern intelligent transportation systems. Some existing methods applied to remote sensing images do not perform well in terms of accuracy due to three primary constraints. First, high-resolution satellite remote sensing images can be prohibitively expensive, and many open-source datasets are only available in low resolution [4]. The limited effective features extracted from a few pixels can hinder model performance. Second, adverse weather conditions, such as fog and haze, are common in real-world scenarios, which can impact the quality of optical remote sensing images [5]. Finally, modern vehicles are becoming more and more similar in appearance and are difficult to distinguish when classifying their types in remote sensing images.

To address these issues, some methods attempt to use denoising operations to remove haze and obtain clearer images [6]. However, haze- and fog-induced pixel degradation in optical remote sensing images may render dehazing methods ineffective. Other methods attempt to develop a powerful deep-learning algorithm for classification. Luo et al. [7] propose stacking transformer and convolutional neural network (CNN) together to extract both local and global information. However, simply overlaying self-attention operations and convolutional operations can not achieve the desired effect and even interfere with each other. Furthermore, while the transformer can provide the network with a global receptive field, it simultaneously loses the inductive biases that convolutional networks bring, such as translation equivariance and locality, making network training more challenging when the dataset size is not sufficient [8].

In remote sensing image recognition tasks, it is a common practice to improve the model accuracy by aggregating multiscale features, enabling the extraction of multiscale features and global features. This approach is widely used in the field

Manuscript received 13 September 2023; revised 6 November 2023; accepted 21 November 2023. Date of publication 23 November 2023; date of current version 5 December 2023. *(Corresponding author: Jianjun Yuan.)*

The authors are with the College of Artificial Intelligence, Southwest University, Chongqing 400715, China (e-mail: jianjuny@sina.com).

Digital Object Identifier 10.1109/TGRS.2023.3336546

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.

of remote sensing image scene recognition [9], [10] [11]. However, this method proves to be less effective in the identification of small objects within remote sensing images. On one hand, features at different levels have varying dimensions, and any attempt to adjust the size of feature maps will inevitably lead to information loss. In addition, not all levels of features are valuable, and the repeated use of multiple hierarchical features may increase the interference of low-quality or redundant features on the model. On the other hand, in small object recognition, low resolution has become an unavoidable issue, which increases the difficulty of feature extraction.

Inspired by the aforementioned method [7], we develop a powerful and effective end-to-end neural network model, named Dense Generative Net (DGNet), for vehicle classification under foggy weather. We build a global effective receptive field (ERF). CNN architecture is utilized to achieve similar effects of the transformer. This approach not only attains an effective global receptive field but also preserves all the inductive biases of CNN, making the model easier to train compared with transformer. It also avoids issues such as feature alignment that arise from using the aggregation of multiscale features method. Furthermore, we creatively integrate generative tasks into the classification task to tackle the challenge of low image quality in remote sensing images. In addition, we design an innovative generation module based on SR GAN, which generates richer feature information and, to some extent, fundamentally addresses the issue of poor image clarity in remote sensing. Finally, for achieving better effects of classification, a novel dense feature fusion net is designed to integrate original feature information and generate richer feature information. In summary, this article makes the following contributions.

1) The proposed DGNet provides a novel approach to recognize vehicle types for remote sensing images. It effectively addresses the challenges mentioned earlier in the current vehicle recognition from remote sensing images. Furthermore, the parameters of DGNet are very small, approximately 4.5M.
2) The proposed generation layer innovatively integrates the SR GAN into image recognition tasks, effectively addressing the issue of low resolution in remote sensing images. In addition, a dense feature fusion layer is developed to integrate the output of the generation layer into the global feature information extracted from original images for enhancing accuracy.
3) A comprehensive analysis of vehicle-type classification is conducted on remote sensing images collected from five different regions and achieved the best results to date, with a classification accuracy of 99%. Furthermore, we simulate mild, moderate, and heavy foggy weather conditions to evaluate the performance of the proposed model under three different foggy conditions. The evaluation results show that the classification accuracy still surpasses other state-of-the-art methods.

The remainder of this article is organized as follows. Section II discusses the related work. Section III describes the proposed network architecture in detail. Experimental results and analysis are in Section IV. The conclusion is given in Section V.

II. Related Work

In the past few decades, various methods for vehicle-type classification have been proposed and have been successfully applied in the fields of transportation and military. In this article, we mainly discuss deep learning-based approaches. The related models are introduced as follows.

Deep learning has been widely applied in various visual fields and has achieved remarkable results [12]. The classical CNN architecture achieves feature extraction from images by stacking multiple convolutional and pooling layers and using a backpropagation algorithm for training, and finally generates the probability of each vehicle type [13]. However, the success of vision transformer (VIT) [14] has greatly challenged the status of CNN [15], [16], as VIT has demonstrated superior performance in many visual tasks. Meanwhile, some researchers have attempted to combine convolutional layers with attention mechanisms to establish global and local correlations, achieving promising results [7]. Why is VIT more powerful than traditional CNNs? Some researchers believe that this is due to the use of multi-head self-attention (MHSA) in VIT [17]. However, recent works have proposed different views. Liu et al. [18] proposed a modernized pure CNN network called ConvNeXt, which achieved comparable or even superior results than VIT. Ding et al. [19] suggest that VIT's excellent performance is more attributed to its globally ERF rather than its self-attention mechanisms.

Many methods were proposed on SR [20]. Wang et al. [21] proposed an enhanced super-resolution generative adversarial network (ESRGAN) model without the batch normalization layer, which can generate more realistic and detailed textures. Wang et al. [22] introduced a high-order degradation modeling process to better simulate complex real-world degradations and employed an U-Net discriminator with spectral normalization to increase discriminator capability and stabilize the training dynamics. It achieved an excellent performance on various real-world datasets. Rabbi et al. [23] applied ESRGAN to small- and medium-sized object detection in remote sensing images and proposed an efficient model that integrates ESRGAN, edge-enhancement network (EEN), and faster region-based convolutional neural network (FRCNN).

Luo et al. [7] proposed a deep learning algorithm called Dense-transformer in transformer (TNT) for vehicle-type classification tasks based on remote sensing images. It utilizes a combination of stacked Transformers and CNNs to extract both local and global information. This method achieved better performance compared with other deep learning models. However, it did not address the fundamental issue of low-resolution in remote sensing images. It only slightly improved the feature extraction capability of the network. Therefore, while this method showed improved accuracy compared with previous approaches, it still falls short of being satisfactory.

III. Methodology

The main purpose of this article is to establish a powerful and effective end-to-end vehicle-type classification network for remote sensing images under different foggy weather conditions. The proposed network is mainly divided into three parts: feature layer, generation layer, and dense feature fusion

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.

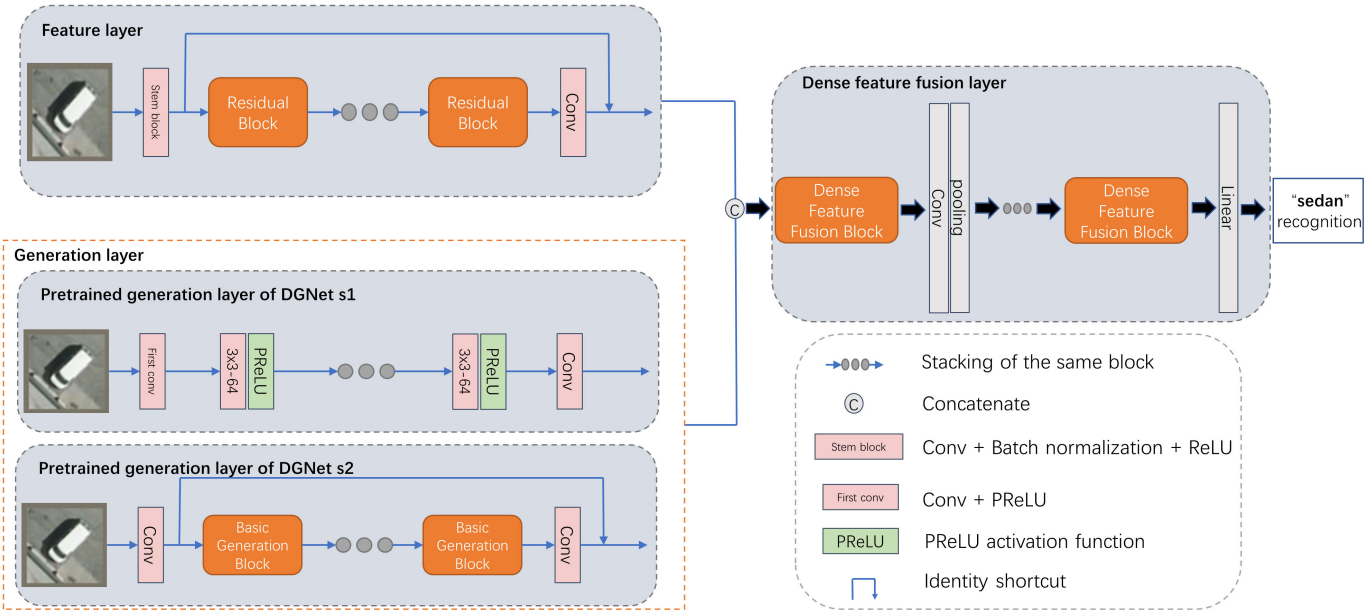


Fig. 1. Overall architecture of DGNet.

layer. Fig. 1 displays the overall architecture of the proposed DGNet.

A. Feature Layer: CNN With Large Kernel

Since VGGNet, stacking small kernel-sized (i.e., 3×3) convolutional layers seems to have become the default standard for CNN network design. The reason is that three 3×3 convolutions can achieve the same receptive field as a single 7×7 convolution, while having far fewer parameters. However, some studies have indicated that in practical applications, increasing the depth of small-kernel models does not significantly increase the ERF, while large-kernel models have a very large ERF. The ERF is proportional to $K(L)^{1/2}$, where K is the kernel size, and L is the network depth [24]. Therefore, we use a larger convolutional kernel (7×7) in this layer to obtain a larger global ERF.

To reduce the significant increase in the number of parameters that comes with enlarging the convolutional kernel, we employ depth-wise convolutions, which have been proven effective in practice [25]. The final total number of parameters in our proposed model is quite promising.

Meanwhile, for convolutional layers with a large kernel size, the identity shortcut is crucial [19]. Without the identity shortcut, a large kernel CNN have difficulty for capturing local details. Shortcuts generate a combination of receptive fields. Without them, the receptive field is singular and huge, making it difficult to capture small features [26].

According to the above, the feature layer in our proposed model is mainly composed of several residual blocks, which includes a 7×7 deep convolution and two commonly used 3×3 convolutions. The important role of the subsequent 3×3 convolution is to compensate for the problem of the deep convolution not integrating interchannel information. The specific structure is shown in Fig. 2.

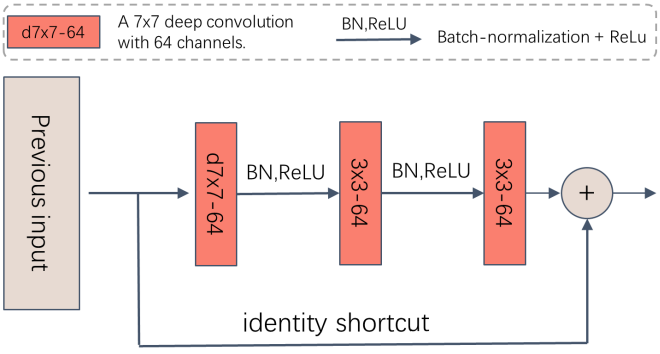


Fig. 2. Basic block of feature layer.

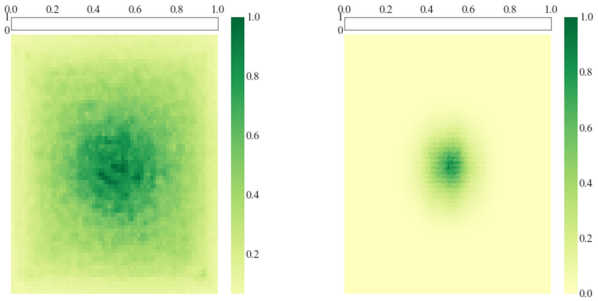


Fig. 3. (Left) ERF of DGNet. (Right) ERF of ResNet101. The ERF of DGNet covers the entire image, with a greater emphasis on the central region.

In Fig. 3, we visualize the ERF range of DGNet after completion of training and compare with ResNet101. The darkness of the colors represents the gradient magnitude of each pixel in an original image with respect to the feature map in the final layer. When the gradient is 0, it means that the pixel will not affect the final result regardless of how it changes, indicating that it is not within the ERF.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.

From the figure, it is evident that DGNet, in comparison with other models, possesses a global ERF, enabling our model to capture globally effective features. In contrast, transformer-based models, while also having a global receptive field, simultaneously lose some of the inductive biases inherent to CNNs, such as translation equivariance and locality. This hinders their ability to generalize well, especially when the dataset is insufficiently large. Furthermore, in some remote sensing tasks, there often is not a vast dataset that matches the scale required by transformers.

B. Generation Layer

Vehicles in remote sensing images are often small targets consisting of only a few dozen pixels and are viewed only from a top–down perspective. In such cases, the distinction boundaries between different vehicle categories, which are already ambiguous, become even more difficult to extract. Therefore, we propose to use an SR generation network to generate richer information based on an original input image, in order to extract some key features. By using this approach, we have managed to address the most significant issue of low resolution.

After carefully comparing the models of a different literature, the generator of real-ESRGAN is selected as a variant in the proposed model. In addition, considering the issue of parameter quantity, we also provide two options here. In this article, the main focus is on the generator of the real-ESRGAN model, while the other components, such as the discriminator, are not directly used. These components will be trained during pretraining. Therefore, we only introduce the variant of the generator used in our proposed model.

Another important point to emphasize is that thanks to the high-order degradation modeling process proposed in [22], and real-ESRGAN has good generalization ability. In order to demonstrate that our proposed model performs well in various situations and locations, we do not pretrain the generation layer specifically on the Cars Overhead With Context (COWC) dataset that we use. Instead, we directly adopt the weights from [22], and these weights are frozen during the training of the entire model. This means that the proposed generation layer is not trained on the training set from start to end.

1) Generation Layer Based on Real-ESRGAN: The generation layer based on real-ESRGAN is a key component of our proposed model. Real-ESRGAN is a state-of-the-art image SR model that has shown excellent performance in generating high-quality images. In the generation layer of our proposed model, we use real-ESRGAN's generator as the starting point and make several modifications to adapt it to our specific task. Specifically, we remove the upsampling layer because we only need the model to generate richer information, rather than producing high-definition images. The specific network architecture is shown in Fig. 4.

The specific structure of the basic generation block is illustrated in Fig. 5. β refers to residual scaling [27], [28], which scales down the residuals by multiplying a constant between 0 and 1 before adding them to the main path to prevent instability. Compared with previous SR networks,

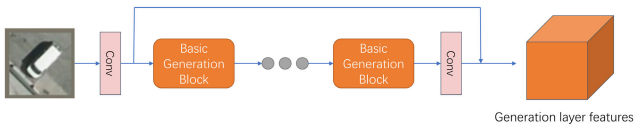


Fig. 4. Overall framework of generation layer.

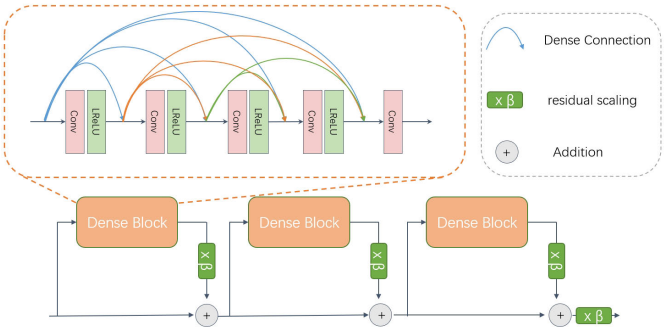


Fig. 5. Basic generation block of generation layer.

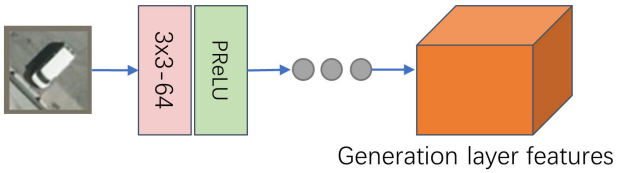


Fig. 6. Generation layer based on realesr-general.

basic generation block removes all batch normalization layers and combines a multilevel residual network with dense connections. These techniques have been proven to be very effective in the field of generating SR images [21]. In a basic block, the output of the last layer is formulated as follows:

$$\mathscr{L}_i^* = \left(\mathrm{Con}\left(\left[\mathscr{L}_0, \mathscr{L}_1, \ldots, \mathscr{L}_{i-1}\right]\right)\right) \times \beta + \mathscr{L}_0 \tag{1}$$

where $[\mathscr{L}_0, \mathscr{L}_1, \ldots, \mathscr{L}_{i-1}]$ is the concatenation result of feature maps from all ahead layers of this block. Con is the final convolutional layer.

2) Generation Layer Based on Realesr-General: Realesr is a simpler and smaller network, with weaker deblur and denoise capabilities, but it has significantly fewer parameters. Similarly, we remove the upsampling part of the generator. The final generation layer adopts a structure similar to that of VGG, as shown in Fig. 6.

C. Dense Feature Fusion Layer

The features obtained from the generation layer are better suited as a complement to the features obtained from the feature layer, rather than being directly suitable for classification tasks (this will be discussed in detail in the experimental section). The results obtained from the generation layer should contain both the original information of an image and generated richer information. However, in reality, this is not always the case. We hope to take both into account to make more accurate judgments. Therefore, we propose a dense feature fusion layer at the end to fuse the features from both sources. Because it is necessary to merge the information passed down

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.

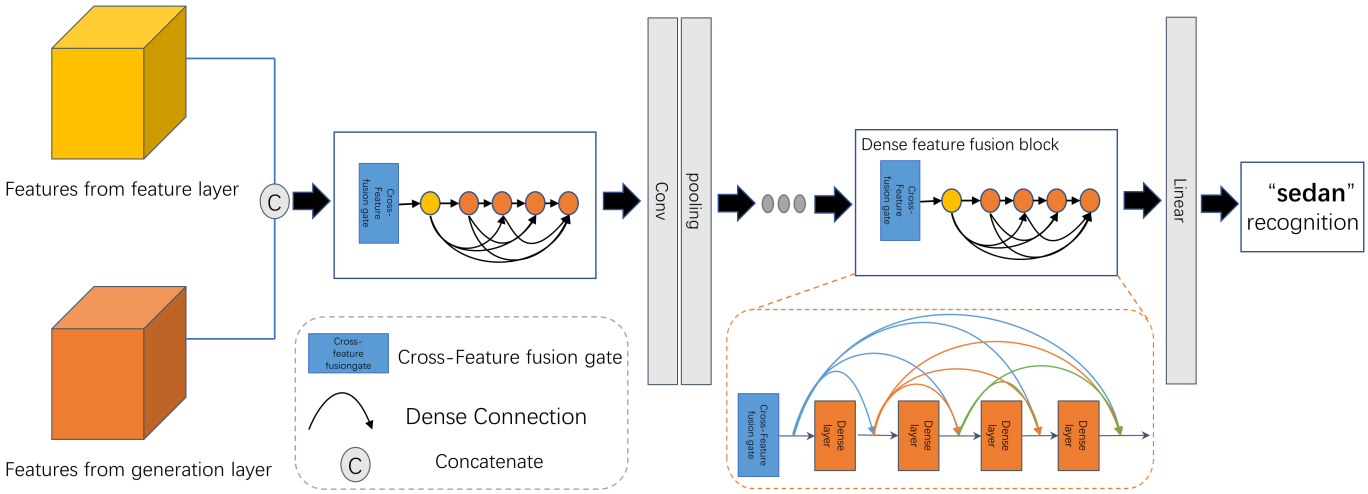


Fig. 7. Dense feature fusion layer.

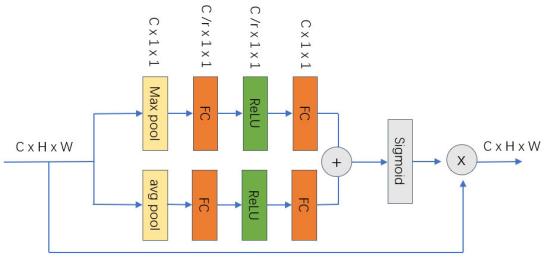


Fig. 8. Cross-feature fusion gate.

from two upstream networks, we utilize dense connections in this module and use channel attention to enable information exchange between channels, achieving the goal of information fusion. The overall dense feature fusion layer is shown in Fig. 7.

1) Cross-Feature Fusion Gate: Cross-feature fusion gate is formulated as follows:

$$\begin{aligned} \tilde{x}_c &= s_c \cdot u_c \\ s &= \sigma\left(w_2\delta(w_1 z_1) + w_2'\delta\left(w_1' z_2\right)\right) \\ z_{1c} &= F_{\text{Avepool}}(u_c) = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j=1}^{W} u_c(i,j) \\ z_{2c} &= F_{\text{Maxpool}}(u_c) = \max(u_c(i,j)) \end{aligned} \tag{2}$$

where u_c represents the cth layer of the feature map output by the upstream network. z_1 and z_2 represent the outputs after applying the average pooling and max pooling layers, respectively. By using this approach, we can squeeze global spatial information into a channel descriptor. In order to fully capture channel-wise dependencies, we pass the squeezed descriptor through two fully connected (FC) layers, represented mathematically by the second expression in (2), where δ denotes the rectified linear unit (ReLU) function, $\mathbf{W}_1 \in \mathbb{R}^{(C/r)\times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C\times(C/r)}$. Finally, we obtain the final output $\tilde{x}_c$ by performing channel-wise multiplication between the feature map u_c and the scalar value s_c. The specific structure is shown in Fig. 8.

Cross-feature fusion gate can establish the connection between different channels of the feature maps from the generation layer and feature layer, selectively enhancing or suppressing features in different channels. Differing from feature fusion modules in typical models, the cross-feature fusion gate processes distinct feature maps from two images, enabling the model to compare the original features with the more informative generated features. This allows the model to effectively utilize all features provided by two upstream networks. It also helps in avoiding interference from noise and irrelevant features, thereby enhancing the model's robustness, generalization capability, and overall performance.

2) Dense Feature Fusion Block: Dense feature fusion block is the basic building block of the dense feature fusion layer. We adopt dense connections to process the information passed through the cross-feature fusion gate, in order to avoid loss of features. Through this structure, we can ensure that the feature maps obtained from the upstream network are fully utilized, enabling the network to better learn complex feature representations of the samples. The specific structure is shown in Fig. 7.

D. DGNet Overview

In summary, through the three modules of feature layer, generation layer, and dense feature fusion layer, DGNet can achieve an effective global receptive field and make the final classification decision through the fusion of original and generated information. We believe that with this structure, DGNet can extract more truly effective feature information, and through the fusion layer, can more effectively process these information. It can have strong recognition ability even in complex and harsh foggy weather conditions.

IV. Experiment

To evaluate the performance of DGNet, we compare its classification performance with several advanced baselines, including PoolFormer, ConVit, RepLKNet, GCSANet, gated bidirectional network (GBN), FACNN, and Dense-TNT. The classification task mainly involves classifying cars and pickup trucks from images captured by remote sensors in five different regions. We also evaluate the model's classification ability when affected by foggy weather conditions. Finally, we analyze various modules of DGNet through ablation experiments.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.

A. Dataset and Evaluation Criteria

Dataset Description: To perform the classification, we utilize the COWC dataset (http://gdo-datasci.ucllnl.org/cowc/) [29], which is a remote sensing target detection dataset with a resolution of 15 cm/pixel and image dimensions of 64 × 64. To better simulate real-world scenarios, we select remote sensing images from five different regions, including Columbus, Potsdam, Selwyn, Toronto, Utah, and Vaihingen. The dataset consists of approximately 150 000 images. It is worth mentioning that the dataset comprises both color and grayscale images.

Evaluation Criteria: To evaluate the classification results of all models, four metrics are applied to indicate their performance. We first define all classification results in this study as follows.

1) *True Positive (TP):* A sedan is successfully recognized as a sedan.
2) *True Negative (TN):* A pickup is successfully recognized as a pickup.
3) *False Positive (FP):* A pickup is successfully recognized as a sedan.
4) *False Negative (FN):* A sedan is successfully recognized as a pickup.

Four criteria are formulated as follows:

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{3}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{4}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{5}$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}}. \tag{6}$$

The higher the evaluation score is, the better the recognition performance is.

B. Classification Performance

Baseline Settings: The baseline models include PoolFormer, ConVit, RepLKNet, GBN, GCSANET, feature aggregation convolutional neural network (FACNN), and Dense-TNT. PoolFormer [30] is a very popular and powerful transformer model in the visual field. ConVit [31] employs gated positional self-attention to regulate the attention given to the position and content information, thereby introducing a "soft" convolutional inductive bias to the model. This enables the model to combine the strengths of CNN and VIT. RepLKNet [19] utilizes a large 31 × 31 convolutional kernel, achieving comparable or superior results to the Swin Transformer [14] on ImageNet and several typical downstream tasks. Dense-TNT is a specific architecture proposed in [7], which combines transformer and CNN structures and achieves the best classification performance. GBN, GCSANET, and FACNN are advanced algorithms currently used for remote sensing image recognition tasks. FACNN [9] improves the recognition capabilities by aggregating features from multiple hierarchical levels. GBN [10] is a GBN that simultaneously eliminates interfering information while aggregating features from multiple hierarchical levels. GCSANet [11] is a framework based on global context spatial attention and densely connected convolutional networks to extract multiscale global features.

DGNet has two parameter sizes, referred to as $s1$ and $s2$. The main difference between them lies in the generation layer, where $s1$ is based on realesr, and $s2$ is based on real-ESRGAN. The parameter sizes of their generation layers are approximately 1.2M and 16M, respectively. The parameter sizes of the feature layer and dense feature confusion layer are the same, with each being 1.7M and 1.6M, respectively, totaling 3.3M. In the evaluation, we apply Dense-TNT with two parameter sizes ($s12$ and $s24$), PoolFormer with two parameter sizes ($s12$ and $s24$), and ConVit with two parameter sizes (small and base). To ensure a reasonable parameter scale for comparative purposes, we utilize the RepLKNet base in experiments. Among all the models, the ConVit base has the largest parameter size, at around 82M, followed closely by the RepLKNet base with a parameter size of approximately 75 million. The parameter size of ConVit small is 26 million, slightly higher than the parameter sizes of Dense-TNT s24 and PoolFormer $s24$, both of which are around 21 million. The parameter counts of Dense-TNT $s12$ and PoolFormer $s12$ are both 12M. The parameter sizes for GBN, GCSANET, and FACNN are approximately 17.2 million, 8.1 million, and 7.3 million, respectively.

Training: The model is trained with 50 epochs on a P100 GPU with a maximum learning rate of $\text{lr} = 2e-3$. Adam optimizer is used with weight decay 0.05. The batch size is 32. The size of training data is 0.8 of the size of the whole dataset. while the size of test data is 0.2 of the size of the whole dataset. All the training data and the test data are randomly selected from the whole set. The loss function utilizes cross-entropy loss. The learning rate update strategy utilizes warm-up and cosine annealing, with the warm-up set to 15. To balance the number of sedans and pickups, we randomly select an equal quantity of both types of images.

1) Classification in Normal Weather Conditions: Table I presents the experimental results, including the classification accuracy and parameter sizes of each model. DGNet $s1$ and DGNet $s2$, respectively, achieve impressive classification accuracies of 97.5% and 99%, surpassing other models significantly in all four evaluation metrics. Moreover, DGNet $s1$ achieves outstanding results with significantly fewer parameters than other models. In addition, Fig. 9 illustrates the confusion matrices of the prediction results for DGNet $s1$ and $s2$. Fig. 10 presents the partial experimental results in the histogram, from which it can be observed that DGNet demonstrates superior performance.

2) Classification in Foggy Conditions: Due to the difficulty in collecting foggy weather datasets and the high cost of updating data labels, as well as the lack of abundant open-source datasets, we used a center point-based synthetic fog method based on the standard optical model to simulate foggy weather data. The specific processed data are briefly described as follows:

$$d = -0.04 \times \sqrt{(i-i_0)^2+(j-j_0)^2} + \sqrt{\max(N_R, N_C)} \tag{7}$$

$$t_d = e^{-\beta \times d} \tag{8}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSIFICATION ACCURACY AND PARAMETER SIZE OF THE COMPARED MODELS ON COWC

	Accuracy	Precision	Recall	F1-score	params
DGNet s1	**0.9749**	0.9715	0.9781	0.9748	**4.50M**
DGNet s2	**0.9900**	0.9923	0.9877	0.9900	19.3M
RepLKNet base	0.9611	0.9912	0.9785	0.9848	75.0M
ConVit small	0.9489	0.9463	0.9515	0.9488	26.0M
ConVit base	0.9507	0.9502	0.9511	0.9506	82.0M
GCSANet	0.9637	0.9625	0.9648	0.9636	8.1M
GBN	0.9492	0.9483	0.9500	0.9491	17.2M
FACNN	0.9477	0.9457	0.9500	0.9478	7.3M
Dense-TNT s12	0.7971	0.8183	0.9399	0.8722	12.0M
Dense-TNT s24	0.8065	0.8211	0.9558	0.8810	21.0M
PoolFormer s12	0.7724	0.7977	0.9424	0.8619	12.0M
PoolFormer s24	0.7819	0.7956	0.9559	0.8672	21.0M

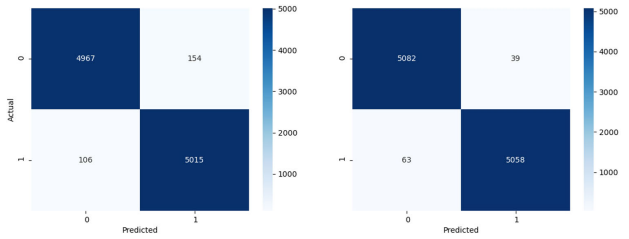


Fig. 9. Confusion matrices of the prediction results for DGNet $s1$ and $s2$ under normal weather conditions. (Left) Prediction results of $s1$. (Right) Prediction results of $s2$.

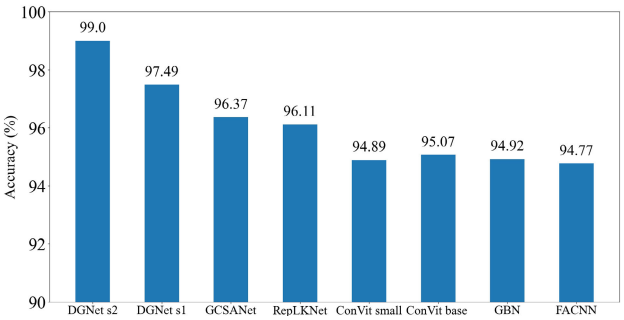


Fig. 10. Accuracy of experimental results under normal weather conditions.

$$V'_{ij} = V_{ij} \times t_d + 0.5 \times (1 - t_d) \tag{9}$$

where i_0 and j_0 represent the fogging center, which is the center of an image, N_R and N_C refer to the number of pixels in each row and each column. β represents the concentration of fog, which is set to 0.08, 0.16, and 0.24 in this experiment to simulate light fog, medium fog, and heavy fog, respectively [7].

Fig. 11 and Table II show the example images and experimental results under different weather conditions. The classification accuracy of the other models is obtained using the same method as in the first part of the foggy weather dataset. The model size and training set remain the same. The evaluation criteria are also the same as in the first part. Fig. 12 displays the accuracy of the experimental results in a histogram. Fig. 13 presents the confusion matrices of the prediction results for DGNet $s1$ and $s2$ under varying degrees of foggy weather conditions.

Due to varying degrees of foggy weather affecting the input data, the accuracy of all models is slightly attenuated. However, even so, DGNet $s1$ and $s2$ still achieve the highest classification accuracy, significantly outperforming the other models. It suggests that DGNet demonstrates excellent generalization capabilities and remains useful when handling affected input data.

C. Ablation Experiment

In this section, we aim to assess the effectiveness of each module within DGNet and evaluate their impact on the overall model performance.

1) Function of Feature Layer: The role of the feature layer is to extract globally effective features from the original image, serving as a benchmark within the entire model framework, while the content generated by the generation layer merely acts as a supplement. In this section, we remove the feature layer from the model and rely solely on the features generated from the generation layer to accomplish the final classification. Experimental results are presented in Table III.

From the table, it can be observed that when using only the generation layer without the feature layer, the model's accuracy significantly decreases. Since the generation layer is designed based on GAN models for generative tasks, it inevitably encounters the common issue of hallucinations that all generative models face. This means that a portion of the features generated by the generation layer is not entirely reliable and needs to be combined with the genuine features extracted by the feature layer to successfully complete the final classification task.

2) Convolutional Kernels of Different Sizes: In Table IV, we compared the performance of using 7×7 convolution and 3×3 convolution in the feature layer.

It should be noted that large convolutional kernels are primarily used within feature layer modules to enhance global feature extraction capabilities. Therefore, in this article, we have chosen to conduct experiments solely on the feature layer module to better validate the effectiveness of the large convolutional kernel model in extracting features from remote sensing images. Specifically, we directly appended a classification head after the feature layer module, and compared their classification accuracy to evaluate the feature extraction performance of both. Apart from the convolutional kernel size, the module structure remains the same. The specific experimental results are presented in Table IV.

As shown in Table IV, the experimental results clearly indicate that using larger convolutional kernels can lead to better classification performance. Larger convolutional kernels provide the model with a broader ERF, which enhances its feature extraction capability.

3) Generation Layers With Different Architectures: In this section, we discuss and analyze the pivotal role that the generation layer plays within the entire model structure. We validate the effectiveness of the generation layer through comparative experiments and delve deeper into understanding its impact on the overall model performance. Specifically, we compare the model's classification performance when we remove the generation layer and when we use different structures for the generation layer.

In the experiment, we first remove the generation layer, retain the feature layer and the dense feature fusion layer,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.

TABLE II

RESULTS OF EXPERIMENTS WITH DATA AFFECTED BY FOG. THE FIRST COLUMN OF THE TABLE REFERS TO DIFFERENT MODELS. THE OTHER COLUMNS SHOW THE ACCURACY OF THOSE MODELS IN LIGHT-FOGGY WEATHER, MEDIUM-FOGGY WEATHER, AND HEAVY-FOGGY WEATHER, RESPECTIVELY

Models	Light-foggy(fog=0.08)				medium-foggy(fog=0.16)				heavy-foggy(fog=0.24)			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
DGNet s1	0.9714	0.9698	0.9731	0.9714	0.9748	0.9728	0.9770	0.9749	0.9734	0.9696	0.9775	0.9735
DGNet s2	**0.9907**	0.9941	0.9873	**0.9907**	**0.9889**	0.9916	0.9861	**0.9883**	**0.9869**	0.9890	0.9848	**0.9869**
RepLKNet base	0.9614	0.9607	0.9619	0.9612	0.9588	0.9566	0.9611	0.9588	0.9532	0.9527	0.9535	0.9530
ConVit small	0.9488	0.9465	0.9511	0.9487	0.9452	0.9415	0.9492	0.9453	0.9435	0.9437	0.9431	0.9433
ConVit base	0.9511	0.9502	0.9519	0.9510	0.9489	0.9486	0.9490	0.9487	0.9466	0.9459	0.9472	0.9465
GCSANet	0.9633	0.9611	0.9656	0.9633	0.9612	0.9561	0.9666	0.9613	0.9578	0.9563	0.9591	0.9576
GBN	0.9489	0.9451	0.9529	0.9490	0.9445	0.9388	0.9507	0.9447	0.9313	0.9366	0.9355	0.9375
FACNN	0.9449	0.9397	0.9507	0.9452	0.9433	0.9411	0.9459	0.9435	0.9400	0.9337	0.9470	0.9403
Dense-TNT s12	0.7907	0.8244	0.9178	0.8671	0.7839	0.7815	0.9510	0.8382	0.7648	0.7748	0.9497	0.8715
Dense-TNT s24	0.7941	8240	0.9215	0.8682	0.7961	0.7934	0.9352	0.8712	0.7692	0.7660	0.9440	0.8787
PoolFormer s12	0.7631	0.7630	0.9289	0.8641	0.7500	0.7469	0.9624	0.8543	0.7371	0.7370	0.9601	0.8469
PoolFormer s24	0.7665	0.7912	0.9243	0.8490	0.7590	0.7630	0.9594	0.8608	0.7535	0.7663	0.9543	0.8635

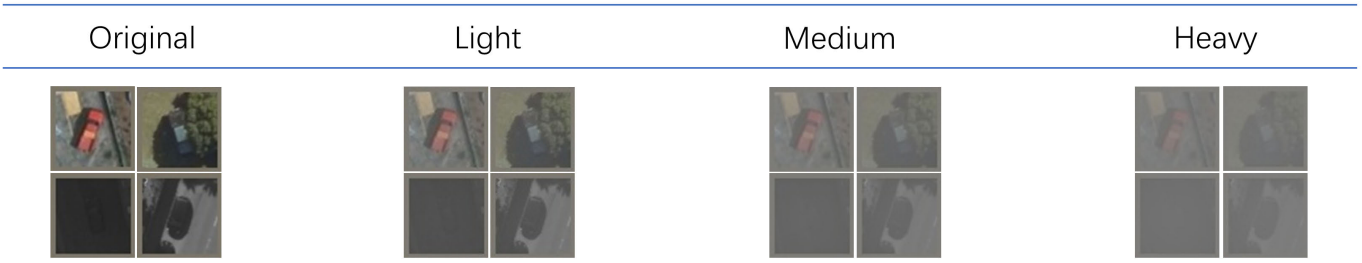


Fig. 11. Experimental images under different weather conditions. The four columns refer to the dataset under normal weather, light fog, medium fog, and heavy fog conditions, respectively.

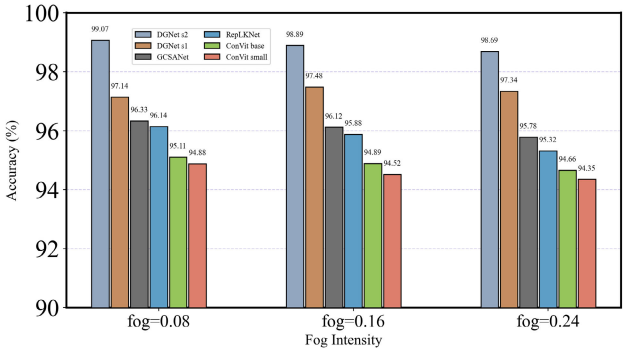


Fig. 12. Accuracy of experimental results under different foggy conditions.

TABLE III

EXPERIMENTAL RESULTS OF DGNET WITHOUT FEATURE LAYER

	Accuracy	Precision	Recall	F1-score
DGNet s2 without feature layer	0.8031	0.7704	0.8635	0.8143
DGNet s1 without feature layer	0.7038	0.7143	0.6792	0.6963
DGNet s2	**0.9900**	0.9923	0.9877	0.9900
DGNet s1	**0.9749**	0.9715	0.9781	0.9748

TABLE IV

EXPERIMENTAL RESULTS WITH DIFFERENT SIZES OF CONVOLUTION KERNELS

	Accuracy	Precision	Recall	F1-score
Feature Layer(3×3)	0.9267	0.9270	0.9264	0.9267
Feature Layer(7×7)	**0.9685**	0.9657	0.9725	0.9690
Feature Layer(3×3)(fog=0.08)	0.9190	0.9072	0.9334	0.9201
Feature Layer(7×7)(fog=0.08)	**0.9674**	0.9608	0.9746	0.9676
Feature Layer(3×3)(fog=0.16)	0.9164	0.9033	0.9326	0.9178
Feature Layer(7×7)(fog=0.16)	**0.9628**	0.9586	0.9674	0.9630
Feature Layer(3×3)(fog=0.24)	0.9143	0.9049	0.9258	0.9153
Feature Layer(7×7)(fog=0.24)	**0.9619**	0.9572	0.9647	0.9609

and keep all other structures the same as in DGNet. The experimental results are shown in Table V.

From Table V, it is evident that removing the generation layer results in a significant drop in classification accuracy.

TABLE V

EXPERIMENTAL RESULTS WHEN THE GENERATION LAYER IS REMOVED

	Accuracy	Precision	Recall	F1-score
DGNet without generationlayer	0.9698	0.9665	0.9738	0.9701
DGNet $s2$	**0.9900**	**0.9923**	**0.9877**	**0.9900**
DGNet $s1$	0.9749	0.9715	0.9781	0.9748

Even when comparing it with the situation in Table IV, where only feature layer is used, and despite the inclusion of a dense feature fusion layer, there is no noticeable improvement in classification performance. The reason for this is that through the feature layer module, we have already extracted sufficiently high-quality original image features. Even with increased network depth and model parameter count, it is a challenging to achieve substantial enhancements in classification performance. The generation layer structure employed by DGNet $s1$ is relatively simpler and provides less supplementary information that is why the improvement in accuracy it brings is not as pronounced as that of DGNet s2.

Due to the limitations imposed by low-resolution remote sensing images, common feature extraction techniques have their constraints. We address this issue to some extent by introducing the generation layer we proposed, which generates additional features as a supplement.

We also compared the experimental results with different architectures of generation layers. The DGNet s1 and DGNet s2 have generation layers based on realesr-general and Real-ESRGAN, respectively. The latter has superior deblur and denoise capabilities. The parameter sizes of the generation layers in DGNet $s1$ and DGNet $s2$ also differ significantly, with the latter having a parameter count as high as 16 million, while the former has only 1.2 million parameters. During

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.

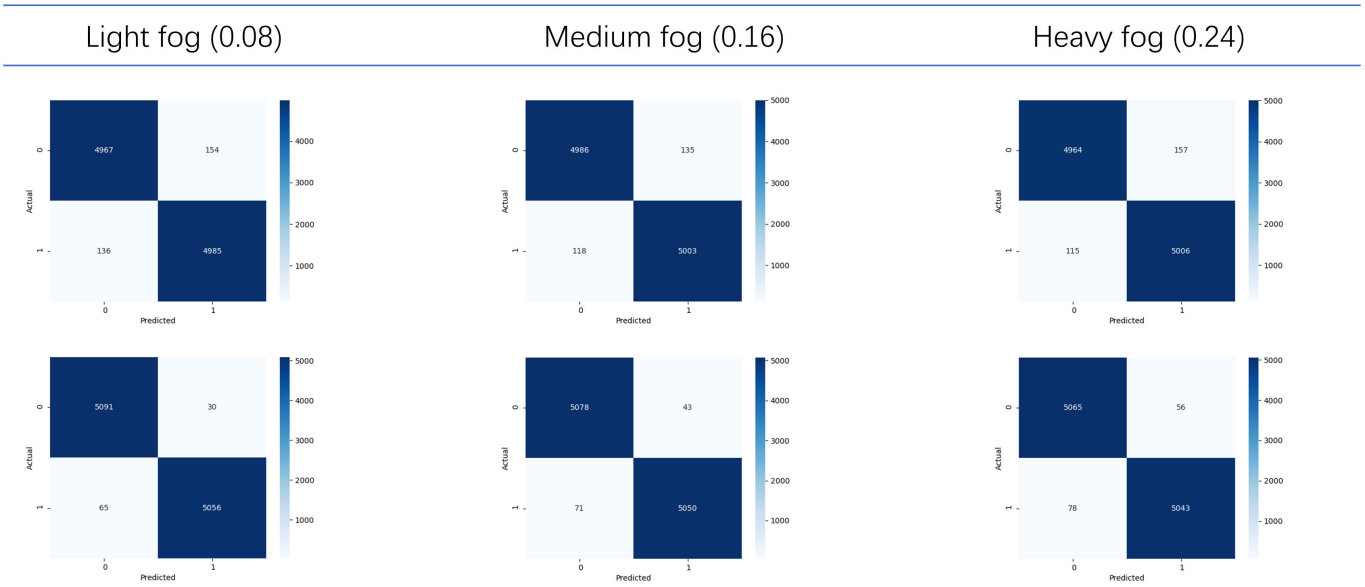


Fig. 13. Confusion matrices under different foggy conditions. The two lines above represent the data of DGNets $s1$ and $s2$, respectively.

TABLE VI
EXPERIMENTAL RESULTS USING DIFFERENT STRUCTURES OF THE GENERATION LAYER

	Accuracy	Precision	Recall	F1-score
DGNet $s1$	0.9747	0.9715	0.9781	0.9748
DGNet $s2$	**0.9900**	**0.9923**	**0.9877**	**0.9900**
DGNet $s1$(fog=0.08)	0.9714	0.9698	0.9731	0.9714
DGNet $s2$(fog=0.08)	0.9907	0.9941	0.9873	0.9907
DGNet $s1$(fog=0.16)	0.9748	0.9728	0.9770	0.9749
DGNet $s2$(fog=0.16)	0.9889	0.9916	0.9861	0.9883
DGNet $s1$(fog=0.24)	0.9734	0.9715	0.9775	0.9735
DGNet $s2$(fog=0.24)	0.9869	0.9890	0.9848	0.9869

TABLE VII
EXPERIMENTAL RESULTS WHEN DENSE FEATURE FUSION LAYER IS REMOVED

	Accuracy	Precision	Recall	F1-score
DGNet $s1$without feature fusion	0.9675	0.9616	0.9742	0.9678
DGNet$s1$	0.9749	0.9715	0.9781	0.9748
DGNet $s2$without feature fusion	0.9674	0.9716	0.9638	0.9677
DGNet $s2$	0.9900	0.9923	0.9877	0.9900

training, the parameters of the generation layer were frozen throughout the entire process.

The experimental results are shown in Table VI. When the parameter size of the generation layer is larger, and the generation quality is better, the overall performance of the model is significantly improved. When the parameter size of the generation layer is increased by more than ten times, the classification accuracy roughly increases by 1.5% points. For SR networks, they perform a low-level semantic unit reconstruction task at the pixel level, while our overall network performs a high-level semantic recognition task. When we use larger models to generate more rich pixels, it has more realistic and better results, naturally providing better feature enhancements for our subsequent classification tasks.

4) Function of Dense Feature Fusion Layer: In this section, we analyze the role of the dense feature fusion layer. Specifically, we remove the feature fusion layer and directly concatenate two feature maps from the upstream for the final classification task. We demonstrate the necessity of the feature fusion module through comparative experimental results.

The specific experimental results are shown in Table VII. It is evident from the data in the table that the use of the dense feature layer significantly improved the model's accuracy. Without a dense feature fusion layer, DGNet struggles to effectively handle features from the two upstream networks, resulting in suboptimal performance for both $s1$ and $s2$. In comparison to the scenario in Table III, where only the feature layer is utilized, the model's accuracy does not improve solely due to the presence of the generation layer. Conversely, in the presence of a dense feature fusion layer, there is a noticeable improvement in accuracy.

V. CONCLUSION

In this article, we introduce a novel classification neural network called DGNet for vehicle-type recognition based on satellite remote sensing imagery. DGNet leverages a feature layer and a generation layer to extract global feature information from the original image and generate richer feature information, respectively. These two sets of features are then fused by using a dense feature fusion layer for classification. Extensive experiments validate the feasibility of DGNet and analyze the effectiveness of each individual module through using the real-world remote sensing dataset collected from multiple regions and diverse environmental conditions. The experimental results indicate that DGNet achieves better recognition performance than other baseline algorithms with fewer parameters. The robustness of DGNet in handling challenging weather conditions, and its ability to maintain high performance is also tested.

REFERENCES

[1] R. Luo et al., "Deep learning approach for long-term prediction of electric vehicle (EV) charging station availability," in *Proc. IEEE Int. Intell. Transp. Syst. Conf. (ITSC)*, Sep. 2021, pp. 3334–3339.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.

[2] R. Luo, Y. Song, L. Huang, Y. Zhang, and R. Su, "AST-GIN: Attribute-augmented spatiotemporal graph informer network for electric vehicle charging station availability forecasting," *Sensors*, vol. 23, no. 4, p. 1975, Feb. 2023.

[3] K. F. Hussain, M. Afifi, and G. Moussa, "A comprehensive study of the effect of spatial resolution and color of digital images on vehicle classification," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 3, pp. 1181–1190, Mar. 2019.

[4] A. X. Wang, C. Tran, N. Desai, D. Lobell, and S. Ermon, "Deep transfer learning for crop yield prediction with remote sensing data," in *Proc. 1st ACM SIGCAS Conf. Comput. Sustain. Societies*, Jun. 2018, pp. 1–5.

[5] P. Barmpoutis, P. Papaioannou, K. Dimitropoulos, and N. Grammalidis, "A review on early forest fire detection systems using optical remote sensing," *Sensors*, vol. 20, no. 22, p. 6442, Nov. 2020.

[6] Q. Wu, W. Ren, and X. Cao, "Learning interleaved cascade of shrinkage fields for joint image dehazing and denoising," *IEEE Trans. Image Process.*, vol. 29, pp. 1788–1801, 2020.

[7] R. Luo et al., "Dense-TNT: Efficient vehicle type classification neural network using satellite imagery," 2022, *arXiv:2209.13500*.

[8] A. Dosovitskiy et al., "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[9] X. Lu, H. Sun, and X. Zheng, "A feature aggregation convolutional neural network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7894–7906, Oct. 2019.

[10] H. Sun, S. Li, X. Zheng, and X. Lu, "Remote sensing scene classification by gated bidirectional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 82–96, Jan. 2020.

[11] W. Chen, S. Ouyang, W. Tong, X. Li, X. Zheng, and L. Wang, "GCSANet: A global context spatial attention deep learning network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1150–1162, 2022.

[12] S. Bera and V. K. Shrivastava, "Analysis of various optimizers on deep convolutional neural network model in the application of hyperspectral remote sensing image classification," *Int. J. Remote Sens.*, vol. 41, no. 7, pp. 2664–2683, Apr. 2020.

[13] Q. Zhang, M. Zhang, T. Chen, Z. Sun, Y. Ma, and B. Yu, "Recent advances in convolutional neural network acceleration," *Neurocomputing*, vol. 323, pp. 37–51, Jan. 2019.

[14] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 10012–10022.

[15] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[16] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[17] S. Paul and P.-Y. Chen, "Vision Transformers are robust learners," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 2, pp. 2071–2081.

[18] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11966–11976.

[19] X. Ding, X. Zhang, J. Han, and G. Ding, "Scaling up your kernels to 31 × 31: Revisiting large kernel design in CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11963–11975.

[20] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[21] X. Wang et al., "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV) Workshops*, 2018, pp. 63–79.

[22] X. Wang, L. Xie, C. Dong, and Y. Shan, "Real-ESRGAN: Training real-world blind super-resolution with pure synthetic data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1905–1914.

[23] J. Rabbi, N. Ray, M. Schubert, S. Chowdhury, and D. Chao, "Small-object detection in remote sensing images with end-to-end edge-enhanced GAN and object detector network," *Remote Sens.*, vol. 12, no. 9, p. 1432, May 2020.

[24] W. Luo, Y. Li, R. Urtasun, and R. Zemel, "Understanding the effective receptive field in deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 1–9.

[25] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1800–1807.

[26] A. Veit, M. J. Wilber, and S. Belongie, "Residual networks behave like ensembles of relatively shallow networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 1–9.

[27] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi, "Inception-v4, inception-ResNet and the impact of residual connections on learning," in *Proc. AAAI Conf. Artif. Intell.*, 2017, vol. 31, no. 1, pp. 4278–4284.

[28] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 1132–1140.

[29] T. N. Mundhenk, G. Konjevod, W. A. Sakla, and K. Boakye, "A large contextual dataset for classification, detection and counting of cars with deep learning," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*, Amsterdam, The Netherlands. Cham, Switzerland: Springer, Oct. 2016, pp. 785–800.

[30] W. Yu et al., "MetaFormer is actually what you need for vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10809–10819.

[31] S. d'Ascoli, H. Touvron, M. L. Leavitt, A. S. Morcos, G. Biroli, and L. Sagun, "ConViT: Improving vision transformers with soft convolutional inductive biases," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 2286–2296.



Jianjun Yuan (Member, IEEE) received the M.Sc. degree in applied mathematics and the Ph.D. degree in instrument science and technology from Chongqing University, Chongqing, China, in 2005 and 2012, respectively.

He joined Southwest University, Chongqing, as a Post-Doctoral Researcher, in 2017. He was the Director of the Department of Data Science and Big Data Technology, and is currently an Associate Professor with the College of Artificial Intelligence, Southwest University. His research interests include machine learning, image processing, deep learning, anomaly detection, and big data analysis.



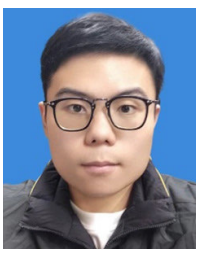
Tong Liu received the bachelor's degree from the School of Mechanical Engineering, University of Shanghai for Science and Technology, Shanghai, China, in 2022. He is currently pursuing the master's degree with the School of Artificial Intelligence, Southwest University, Chongqing, China.

His research interests include deep-learning-based remote sensing image processing.



Haobo Xia received the B.S. degree from the School of Computer Science and Engineering, Sichuan University of Science and Engineering, Yibin, China, in 2022. He is currently pursuing the M.S. degree with the School of Artificial Intelligence, Southwest University, Chongqing, China.

His research interests include remote sensing image processing, computer vision, and deep learning.



Xu Zou received the bachelor's degree from the School of Information Engineering, Qingdao University of Science and Technology, Qingdao, China, in 2021. He is currently studying in computer science and technology with the School of Artificial Intelligence, Southwest University, Chongqing, China.

His main research direction is remote sensing image processing based on deep learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 09:20:11 UTC from IEEE Xplore. Restrictions apply.